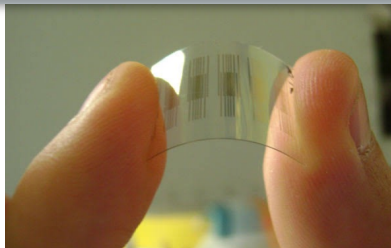
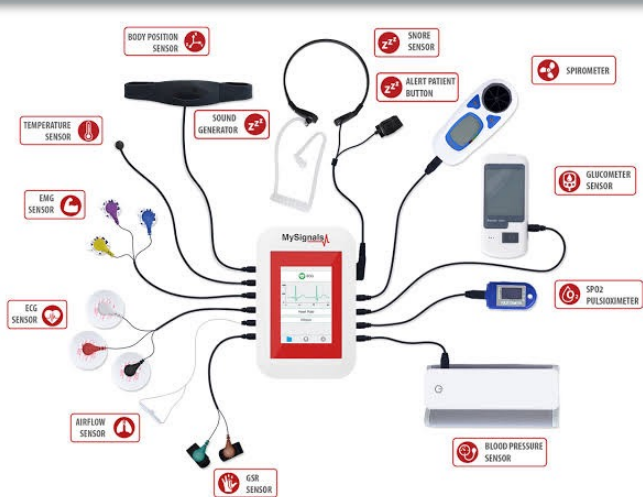
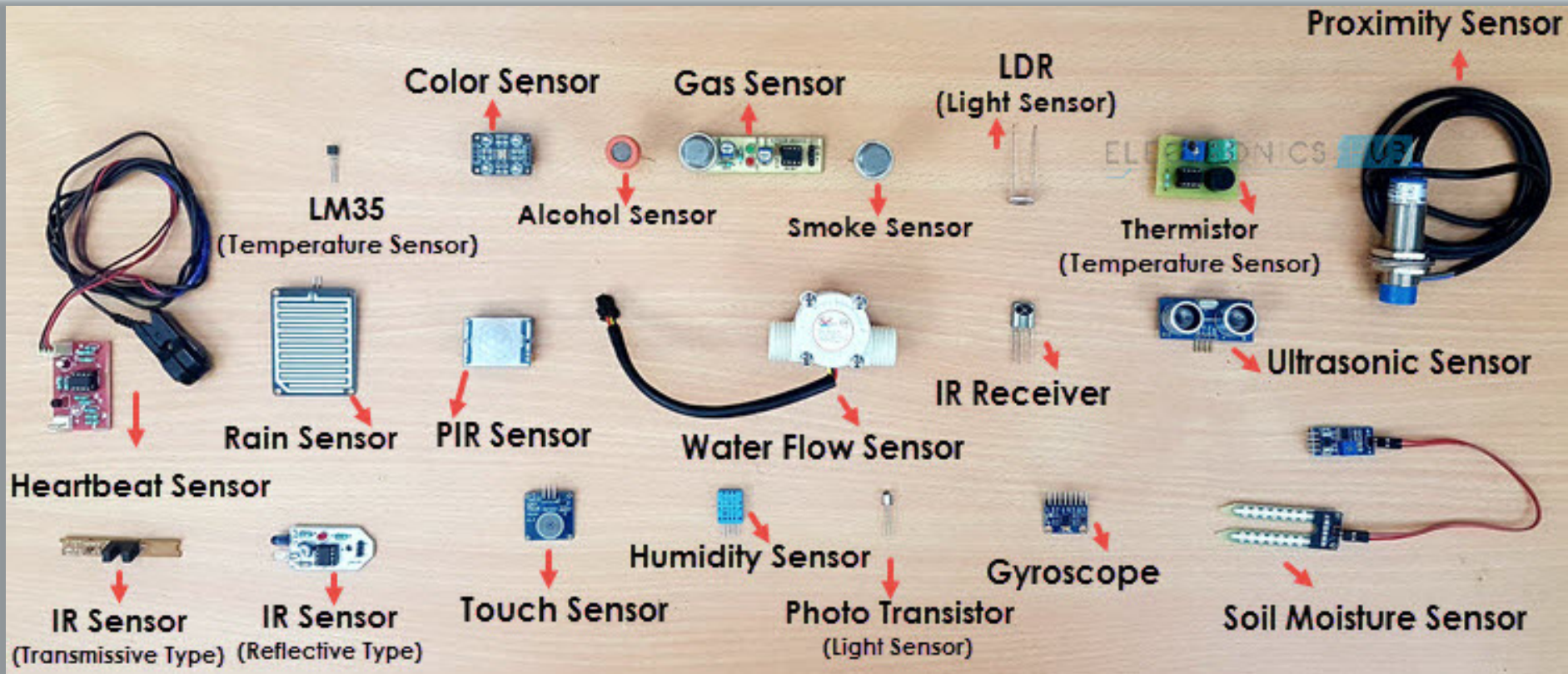


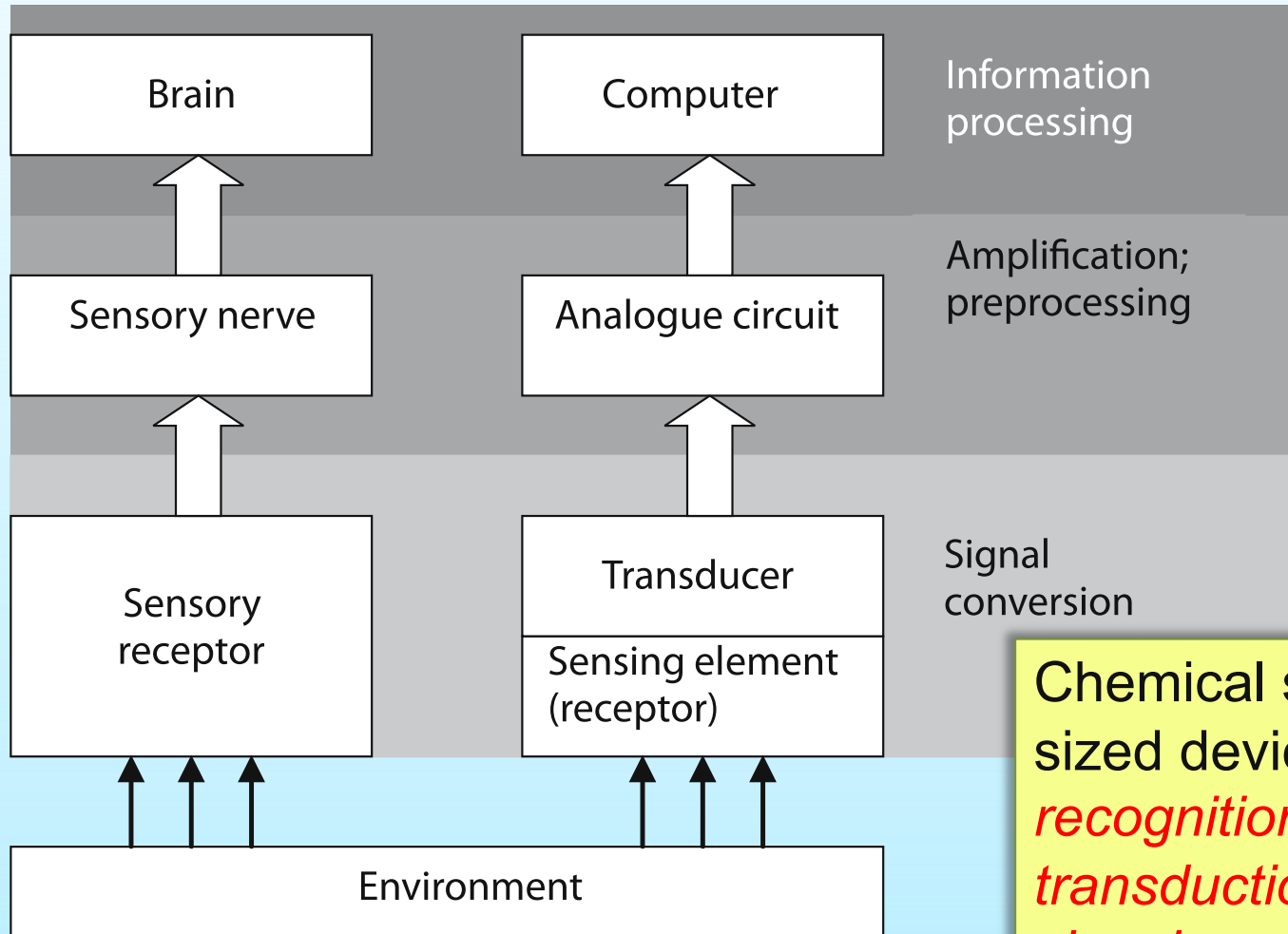
Our world is changing rapidly, and sensors play an important role in this process



Sensors and Detectors



Chemical Sensors



Chemical sensors analyze our environment, i.e. they detect *which substances* are present and in *what quantity*

Chemical sensors can be considered artificial noses or artificial tongues

Chemical sensors are small-sized devices comprising a *recognition element*, a *transduction element*, and a *signal processor* capable of continuously and reversibly reporting a chemical concentration.

Signal and Its Transduction

The interaction between the chemical and the sensor consists of two steps: *recognition* and *amplification*.

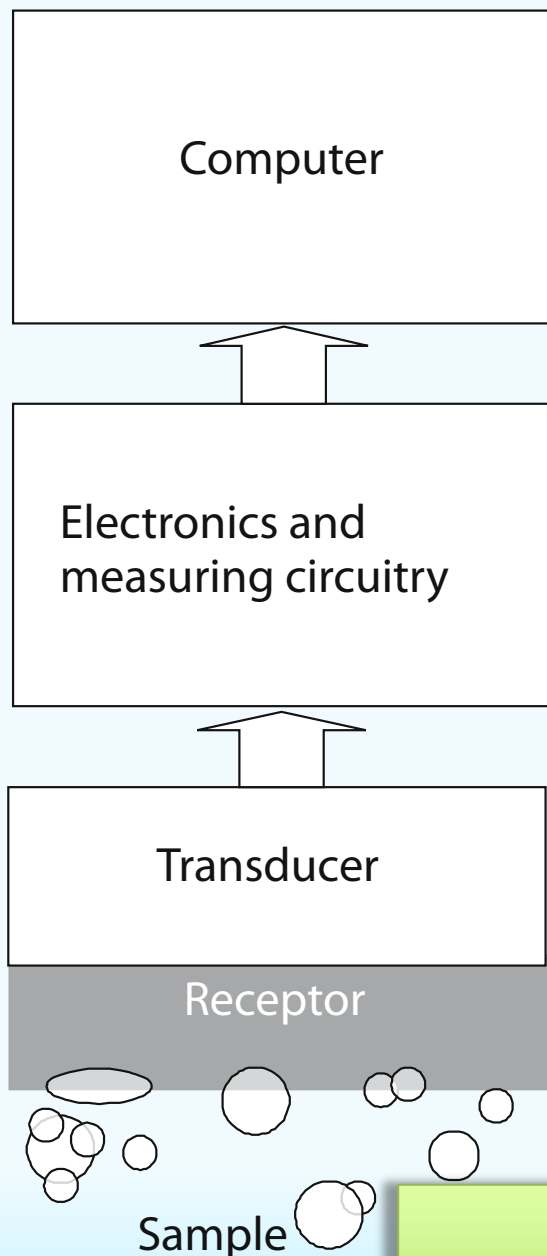
Remember the **pH-meter**, which you have used in the kinetics lab!!



The power density in the primary interaction is very low; any attempt to draw current, i.e., *to obtain a signal*, from such an electrode would result in the distortion of the signal.

Therefore, the source of the signal need to be amplified (using physical *transduction*) to obtain the information in *useful undistorted form*.

Transduction = to transfer, to convert, to transform; in chemical sensing consisting of two basic steps: recognition and amplification



Calculation and display of results, sources of errors, etc.

Amplification, integration, derivation, etc.

Potential difference, current rise, etc.

temperature change, reaction heat, etc.

interaction with sample molecules

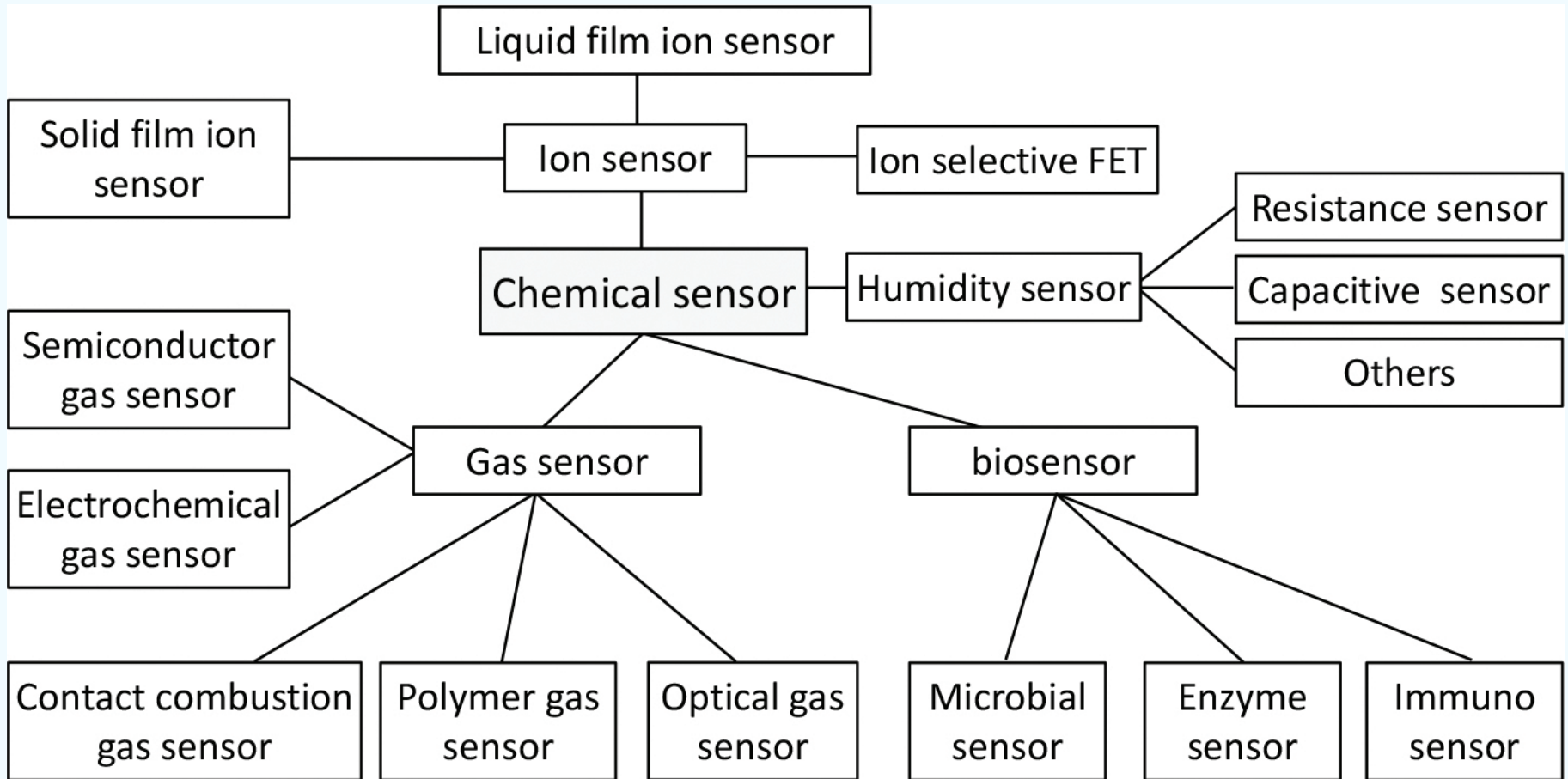
Two components are connected in series: a chemical (*molecular*) *recognition system (receptor)* and a *physicochemical transducer*

The ***receptor interacts with analyte molecules***. As a result, its *physical properties are changed* in such a way that the appending transducer can gain *an electrical signal*

In some cases, one and the same physical object acts *as receptor and as transducer*.

Type of Sensors??
Based on Detection technique, Analyte or ...

The coupling of the chemically selective layer to the physical part of the sensor is very important.



With so many different sensors described in the literature,
which one is the best ?

Criteria to be a Reliable Chemical Sensor

There is no simple answer, because there are so many *different sensing situations and criteria*.

- Respond *quickly*
- Operate *continuously* or at least in repeated cycles
- Be *small*
- Be *cheap*
- Be *specific*, i.e., they should respond exclusively to one analyte, or at least be *selective* to a group of analytes

Ruggedness: A Critical Criteria in Selection of a Sensor

Ruggedness: Ability of the device to maintain its performance specifications even under adverse operating conditions. A *physically rugged* device is able to withstand mechanical shock, vibrations, and other mechanical stresses. *Chemical ruggedness* is related to ***selectivity***; the output of the sensor is unaffected by the unforeseen chemical changes in the operating environment.

Rugged sensors are usually the most commercially successful

The Performance of Chemical Sensors

Sensitivity: change in the measurement signal *per concentration unit* of the analyte..

Selectivity: an expression of whether a sensor responds *selectively to a group of analytes* or even *specifically to a single analyte*.

Detection limit: the *lowest concentration value* which can be detected by the sensor in question, under definite conditions.

Dynamic range: the *concentration range* between the detection limit and the upper limiting concentration.

Resolution: the *lowest concentration difference which can be distinguished* when the composition is varied continuously.

Response time: the time for a sensor to respond *from zero concentration to a step change* in concentration.

Stability: the ability of the sensor to *maintain its performance* for a certain period of time.

Life cycle: the *length of time* over which the sensor will operate.

Selectivity is the single most important general issue in chemical sensing. It determines the usefulness of the sensor for any given application

Low detection limit, or a high sensitivity

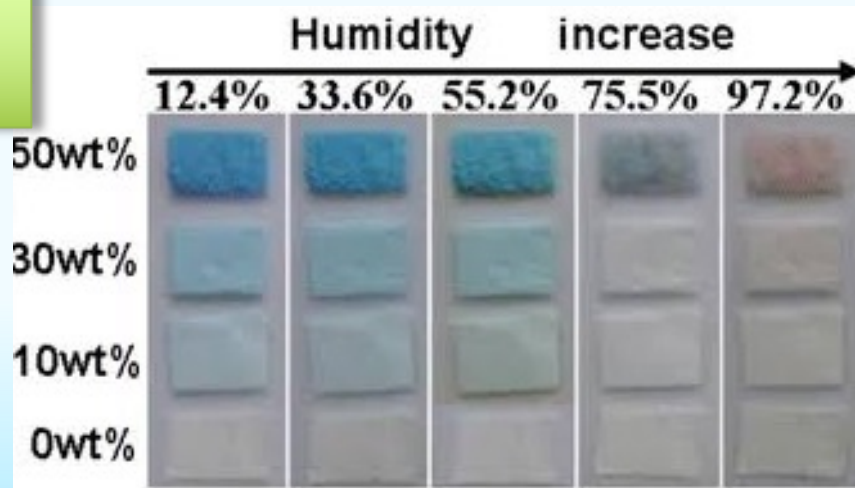
One common example is the measurement of pH with a glass electrode. 10^{-14} M concentration can be measured.



Reversibility in the context of chemical sensing means that the response follows concentration changes, both up and down.

Reversibility follows concentration changes

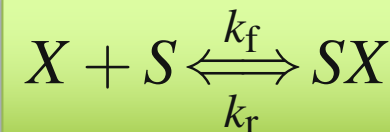
A humidity sensor consists of a cardboard square that is blue when dry and red in humid weather.



Equilibrium, Binding, Response

The “**reading**” of a sensor is generated by the change in some physical parameter, as a result of chemical-sensor interaction; This is called the sensor’s **response**.

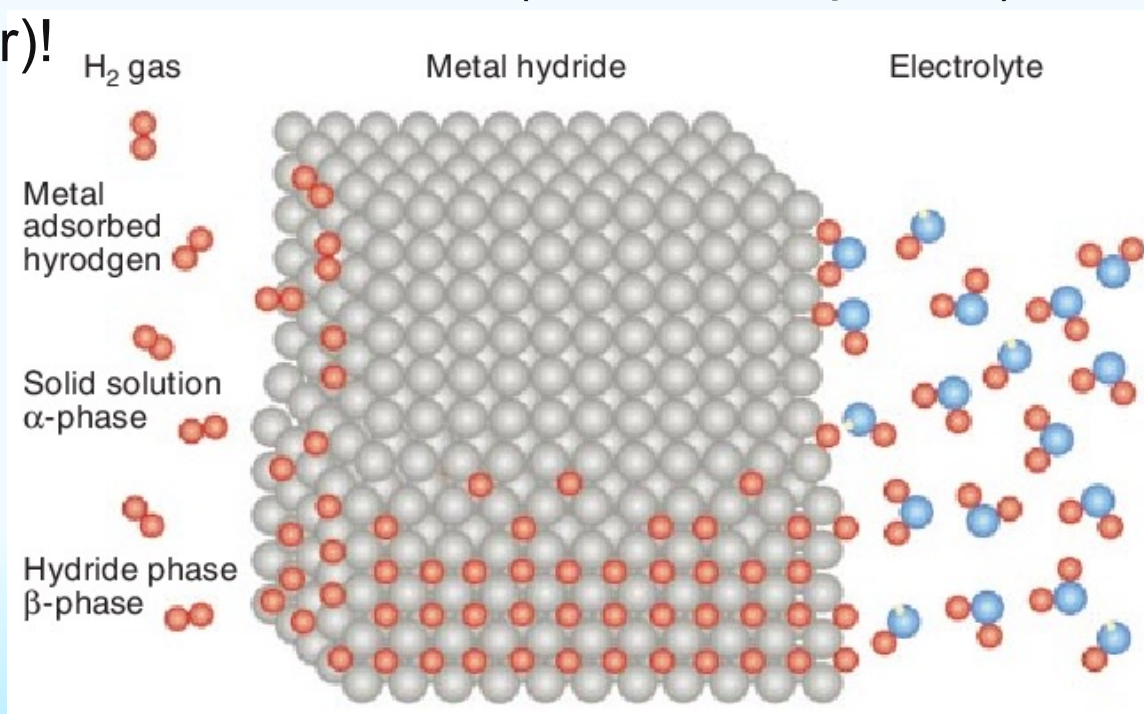
The interaction between the chemical species (X) and sensor (S) can be expressed by the **equilibrium**,



In other words, **binding** of the substrate (chemical species) at **binding site** (of the sensor)!

Surface interaction or adsorption at the surface
Bulk interaction or absorption into sensing element

Interaction equilibria,
Chemical reaction equilibria



Receptor

Receptor layers can respond selectively to a particular substance or to a group of substances. The term **molecular recognition** is used to describe this behavior.

**Receptor-analyte interaction can be classified into
interaction equilibria and chemical reaction**

Processes of interactions are *adsorption* (or absorption), *ion exchange* and *liquid–liquid extraction* (partition equilibrium). Instead of equilibrium, a *chemical reaction* may also become the source of information.

<https://www.bactrack.com/blogs/expert-center/35457349-how-does-a-breathalyzer-work>

Chemical Reactions

pH paper: Determination of $[H^+]$

Measuring $[H^+]$ is typical and very useful in many areas of science.

A pH paper strip is mixture of chemicals; *indicators*, whose colors varies with $[H^+]$.

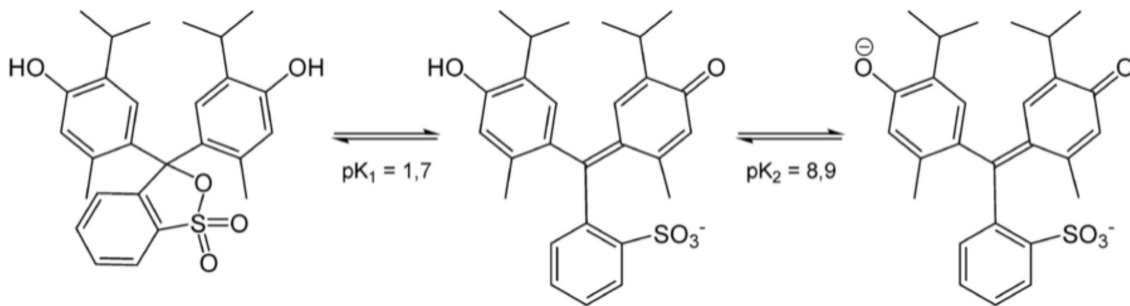


Universal indicator components

Indicator	Low pH colour	Transition pH range	High pH colour
Thymol blue (first transition)	Red	1.2 – 2.8	Yellow
Methyl orange	Red	3.2 – 4.4	Yellow
Methyl red	Red	4.8 – 6.0	Yellow
Bromothymol blue	Yellow	6.0 – 7.6	Blue
Thymol blue (second transition)	Yellow	8.0 – 9.6	Blue
Phenolphthalein	Colourless	8.3 – 10.0	Fuchsia

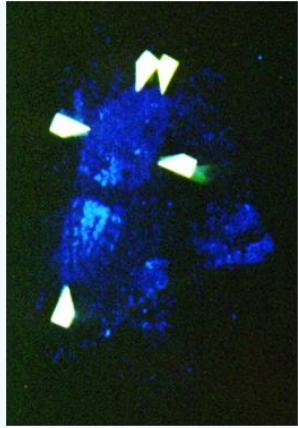
Will a pH paper be called proton sensor?

Formation of new chemical species:



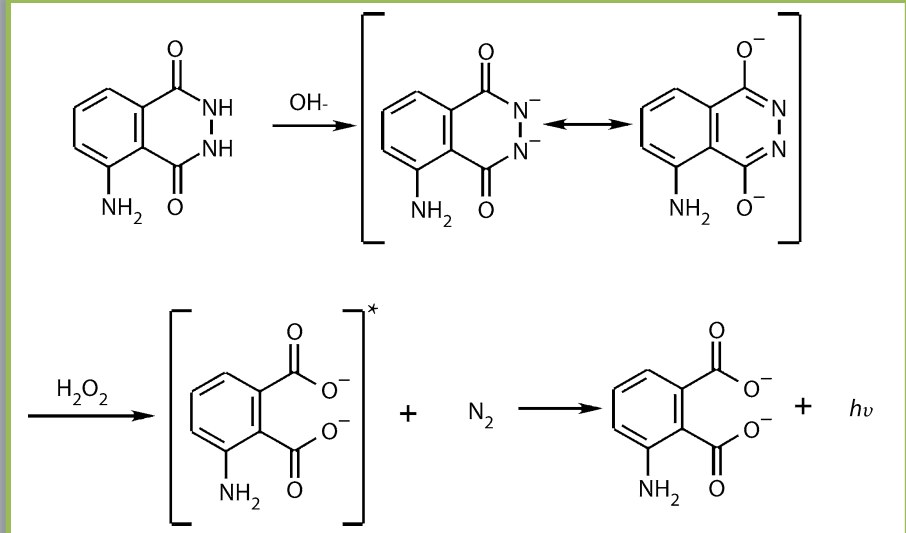
May be?
This is, of course, a chemical analysis; however, no transduction and/or signal processing is involved in the process!

Blood stain: Luminescence



The luminescence techniques can be highly selective as it is possible to distinguish between two different wavelengths (the emission and the excitation one)

The light, or luminescence, emitted in the luminol reaction is thought to result when an oxidizing agent, such as blood, catalyses the oxidation of luminol by hydrogen peroxide in a basic solution.

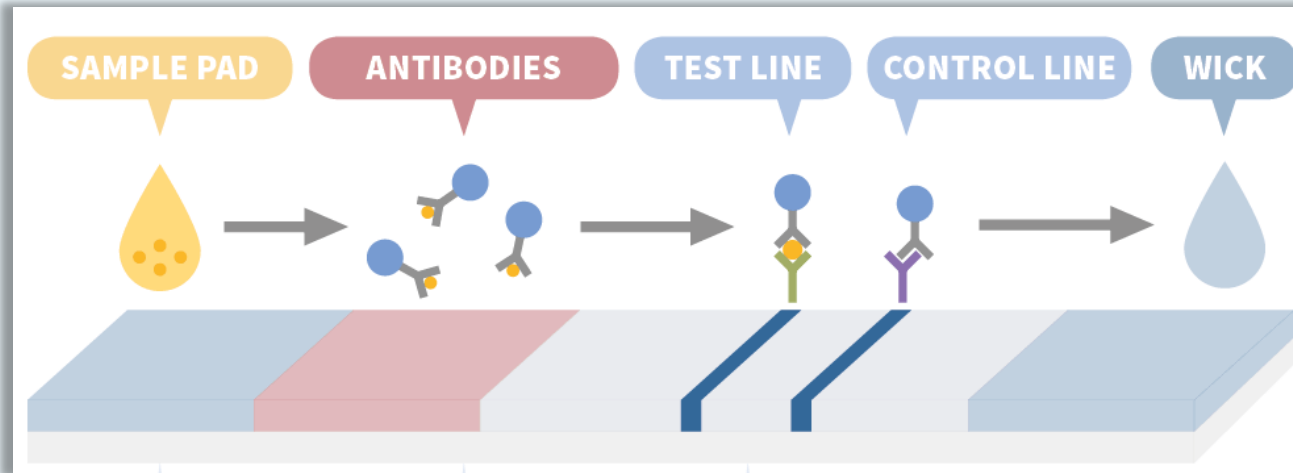


Sensors based on bioluminescence preferably utilize the action of *luciferine*, the luminophore of the firefly. In the *enzyme-catalyzed reaction with oxygen*, **light with a wavelength of 562 nm is emitted**:

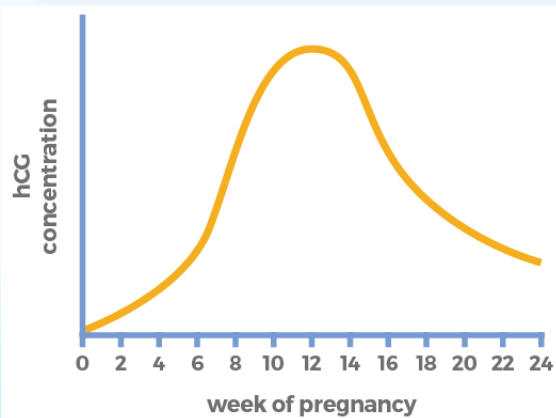
Pregnancy Kit

Pregnancy tests work by detecting the presence of the hormone, *human chorionic gonadotropin* (**hCG**). hCG is a hormone produced by cells in the placenta.

If hCG is present, it *binds to proteins or antibodies*.



These antibodies have an *enzyme attached to them which can participate in the reaction* further along the test strip.



The test results a yes or no!
Does the test provide any duration info?
Can this be called a chemical sensor?
Which type of receptor-analyte interaction is this?
How much accuracy?

Classification following the principles of signal transduction

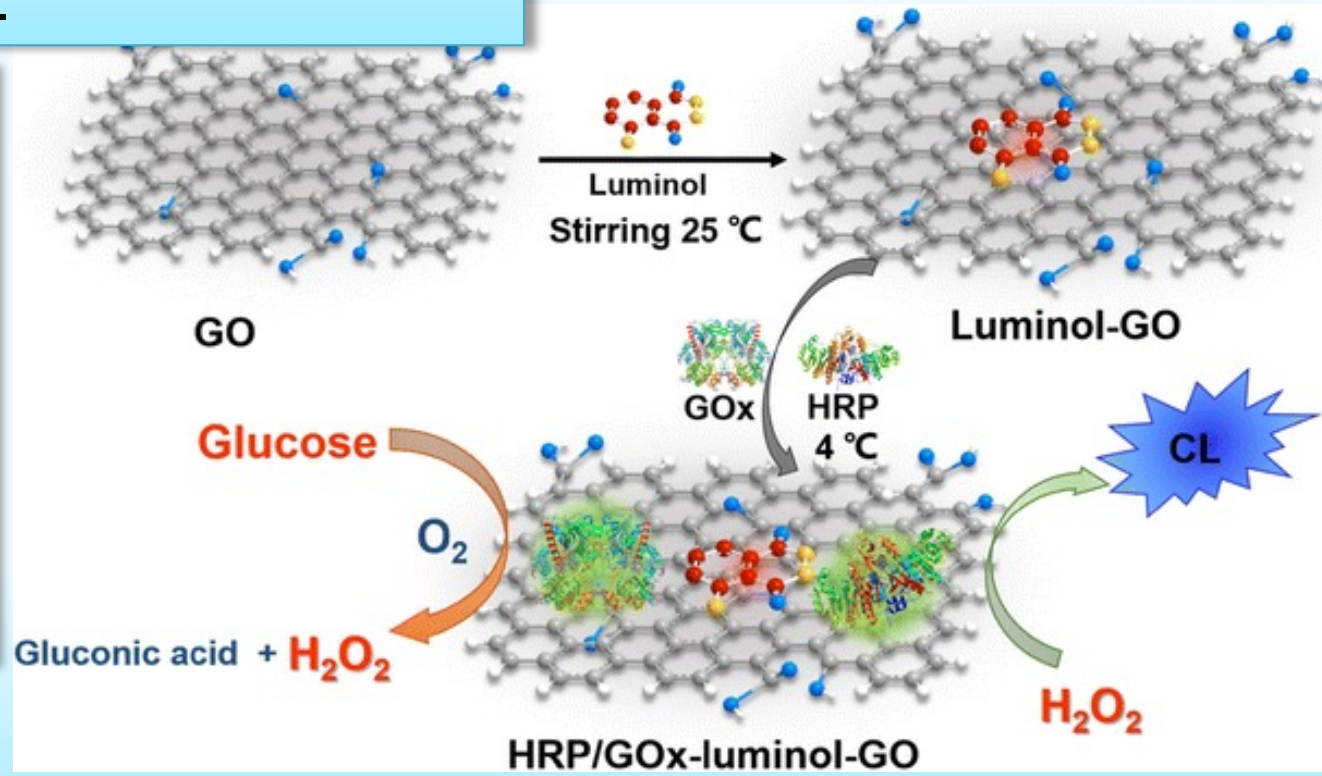
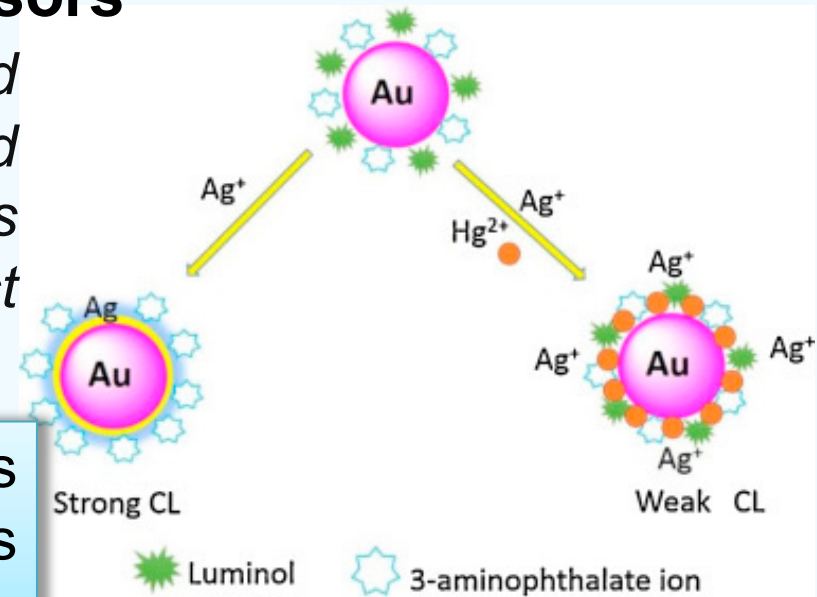
- **Optical sensors**, following absorbance, reflectance, luminescence, fluorescence, refractive index, optothermal effect and light scattering
- **Electrochemical sensors**, among them voltametric and potentiometric devices, chemically sensitized field effect transistor (CHEMFET) and potentiometric solid electrolyte gas sensors
- **Electrical sensors** including those with metal oxide and organic semiconductors as well as electrolytic conductivity sensors
- **Mass sensitive sensors**, i.e. piezo electric devices and those based on surface acoustic waves
- **Magnetic sensors** (mainly for oxygen) based on paramagnetic gas properties
- **Thermometric sensors** based on the measurement of the heat effect of a specific chemical reaction or adsorption which involves the analyte
- **Other sensors**, mainly based on emission or absorption of radiation

Optical Sensors

Conventional spectrophotometric and colorimetric analysis can be performed in an optical sensing format. That makes the optical sensors probably the most universal type of chemical sensors.

Gold nanoparticles with Luminol has been used for the detection of Hg^{2+} ions in aqueous solutions.

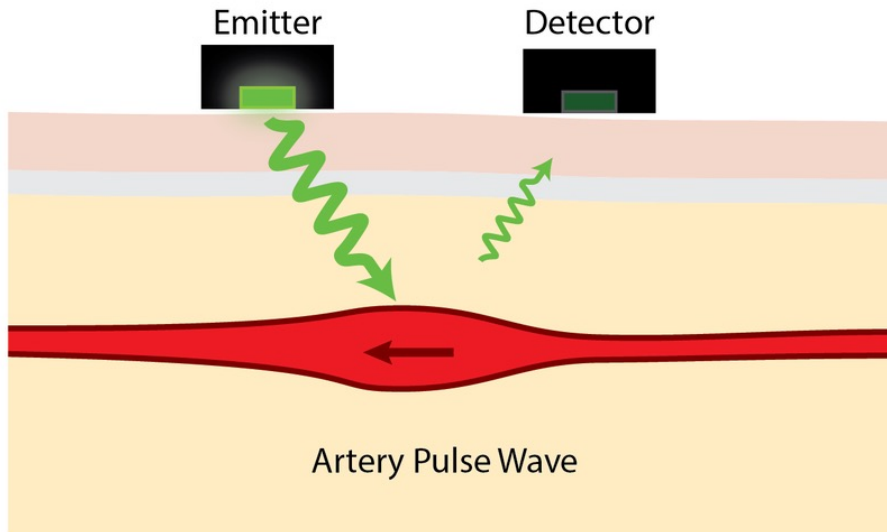
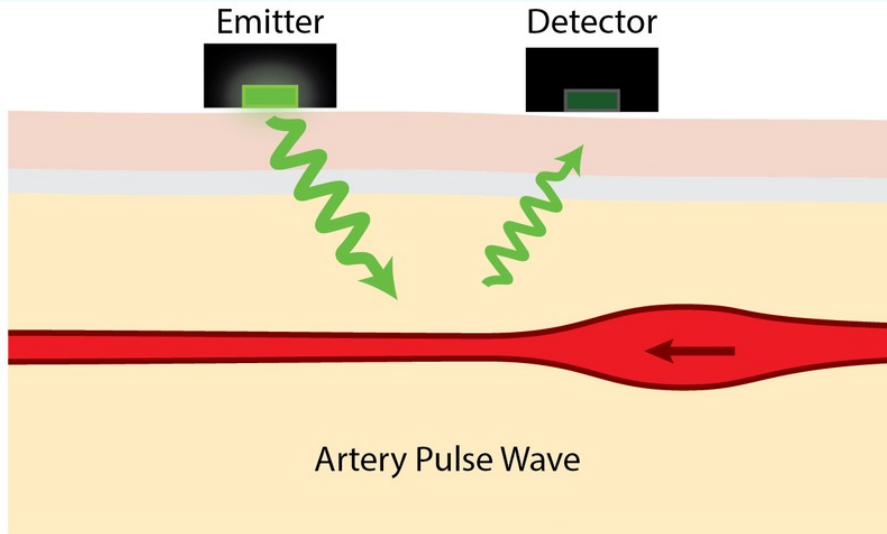
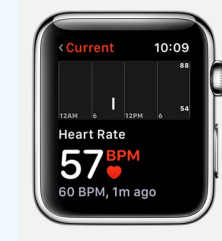
Graphene oxide (GO) catalyzed the luminol- O_2 reaction, which yielded a novel CL. Based on this CL, a new LCL sensor for glucose was constructed.



Optical Sensors

"Photoplethysmography" (PPG), or the process of using light to measure blood flow.

Emission and Detection



The different wavelengths of light from these optical emitters interact differently with the blood flowing through your wrist. When that light refracts (or reflects) off your flowing blood, another sensor in the wearable captures that information.

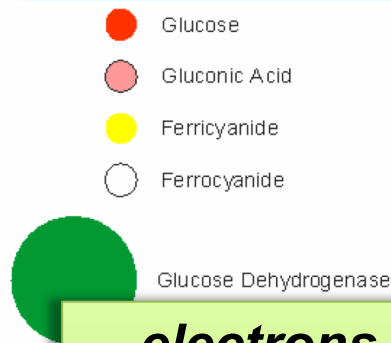
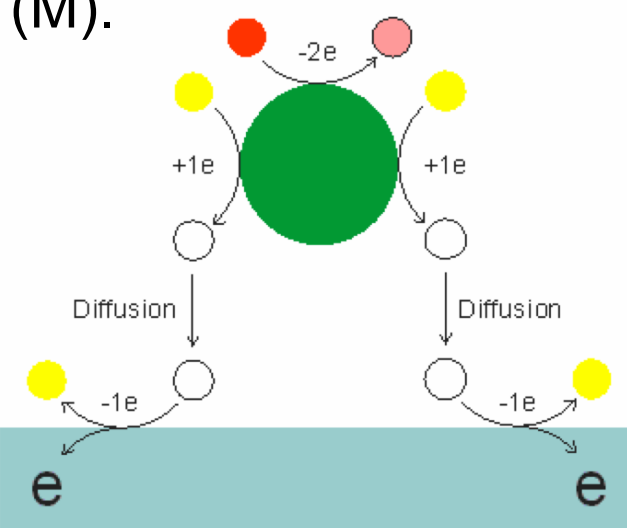
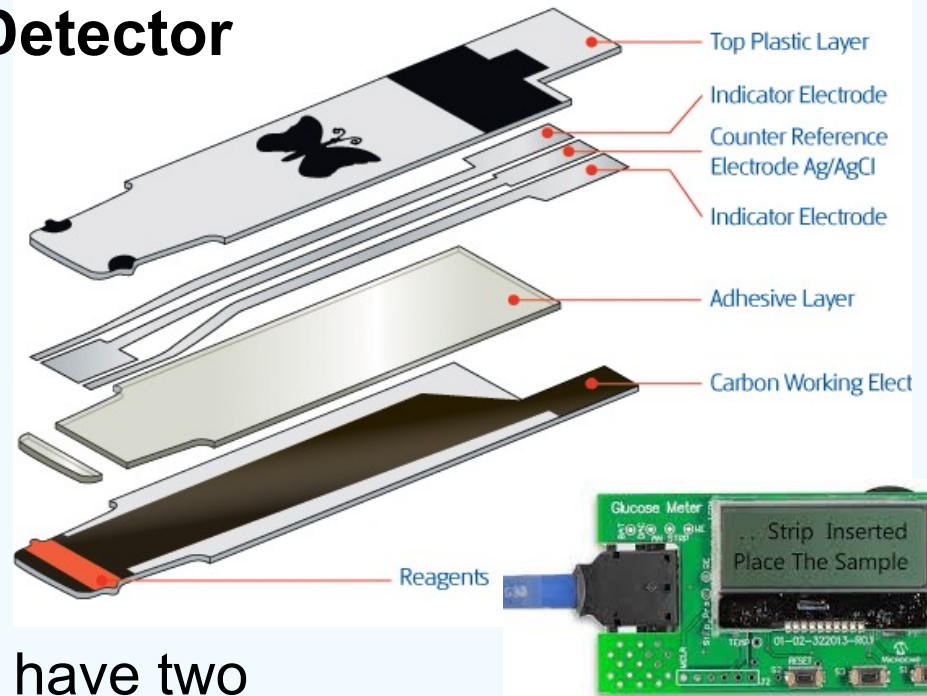
That data can then be processed, along with motion information detected by the device's accelerometer, with algorithms to produce understandable pulse readings.

Glucose Detector



Blood glucose monitors are devices that are used to measure the *concentration of glucose* within a person's blood.

Most basic blood glucose monitors have two separate parts that are required for them to work properly: **a test strip and the meter**. Sample chamber consists of the enzyme and mediator (M).

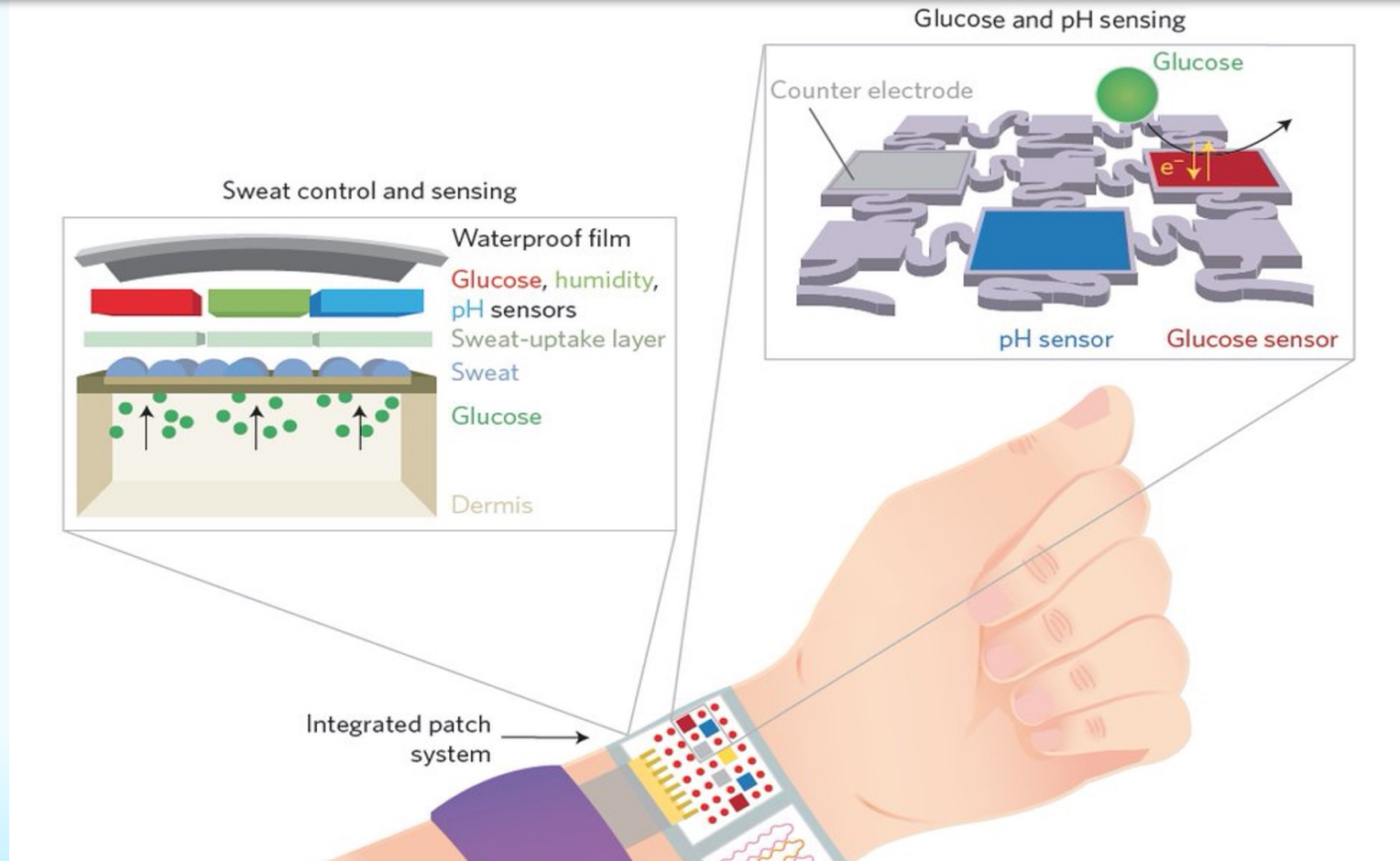


At the electrode surface, M_{red} is oxidized back to M_{ox} and the measured current is used to determine the concentration

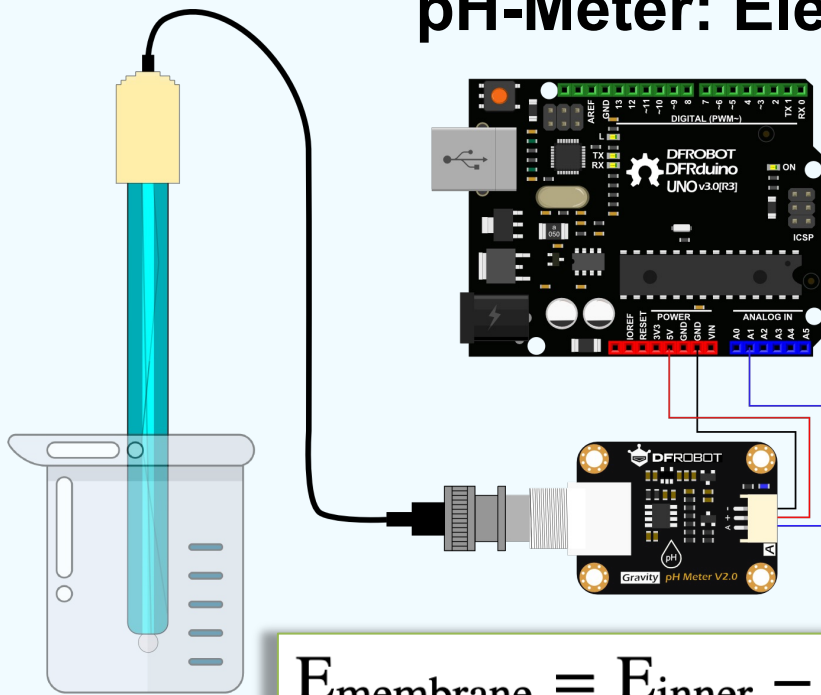
electrons are being transferred between glucose and the electrode; *electrochemical sensor*

Electrochemical Sensors

Charge transport within the transducer part of the sensor, and/or inside the supporting instrumentation, is **electronic**. On the other hand, the charge transport in the sample can be **electronic, ionic, or mixed (electronic/ionic)**.



pH-Meter: Electrochemical Sensor

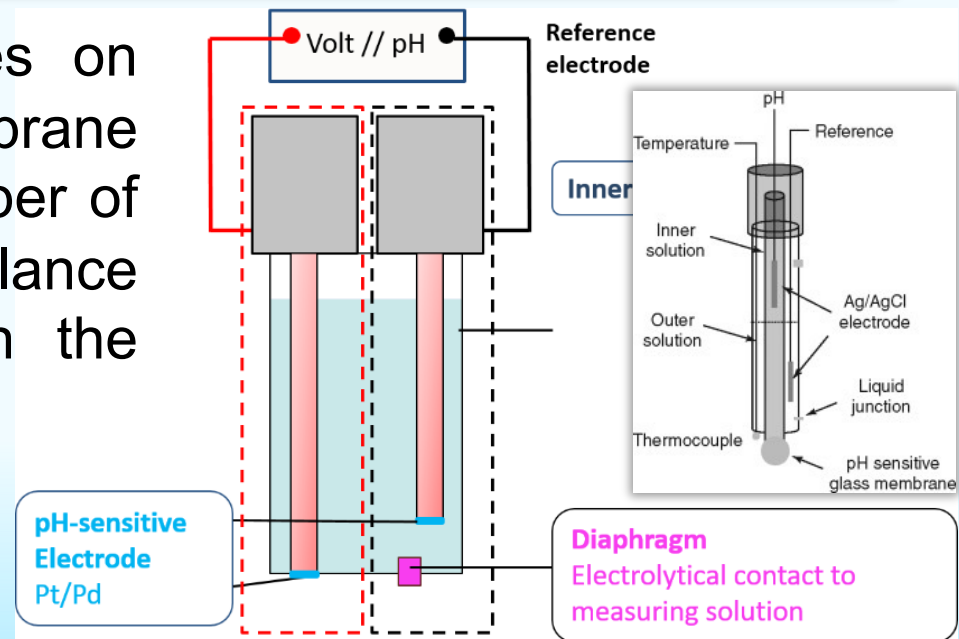


The voltage produced by the test solution is *measured and compared* with the voltage produced by the reference solution. The pair of electrodes in the combinatorial set-up can measure small changes in voltage (or potential difference) in the order of *millivolts*.

$$E_{\text{membrane}} = E_{\text{inner}} - E_{\text{outer}} = 0.0592 \log[(\mathcal{A}_{\text{inner}})/(\mathcal{A}_{\text{outer}})]$$

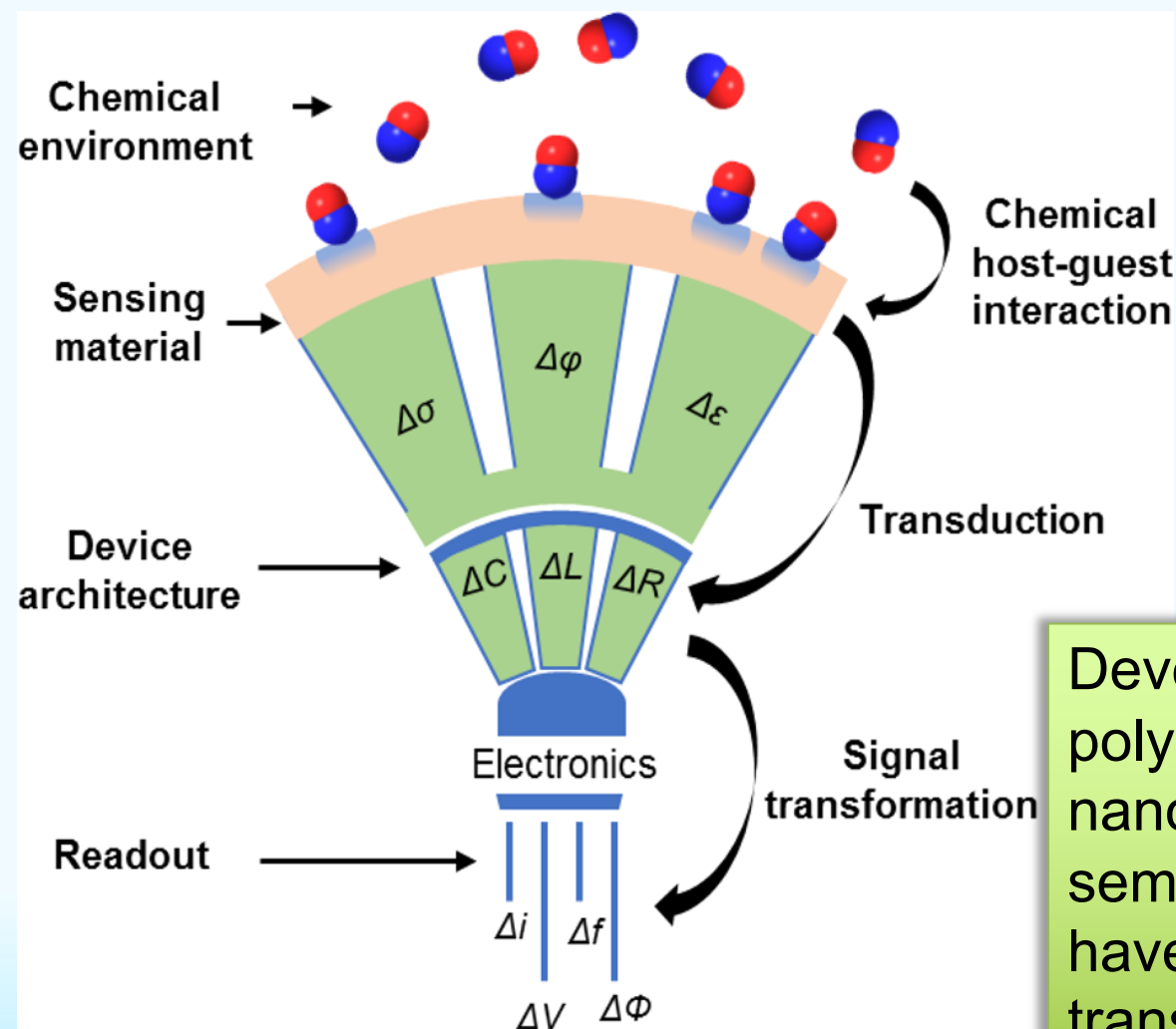
A difference in the H^+ activities on either side of the glass membrane leads to a difference in the number of ion pairs that exist, and an imbalance in the surface charge between the hydrated layers.

This results in a *membrane potential* that is pH dependent.



Electrical Sensor

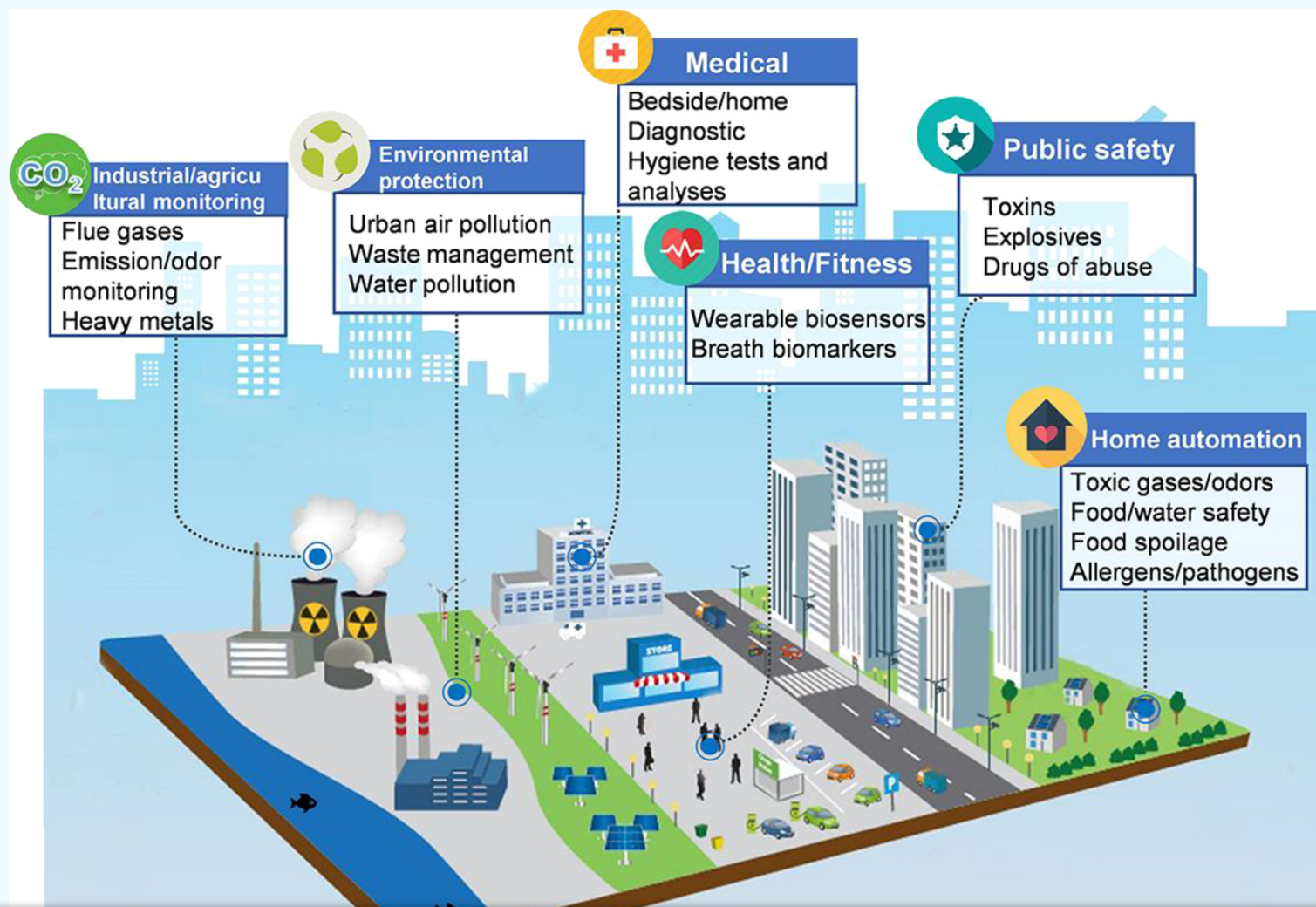
An electrical sensor is a device or integrated circuit that detects a specific physical parameter and converts it to an electrical signal; *current, voltage, resistance, inductance and capacitance* etc.



The sensing material should have a suitable active sites for effective and selective analyte binding, and the **ability to transduce the binding events into a detectable signal**

Development of conductive polymers, fullerenes, carbon nanotubes, and metallic & semiconductive nano-particles, have advanced electrically-transduced chemical sensing.

Electrical Sensor



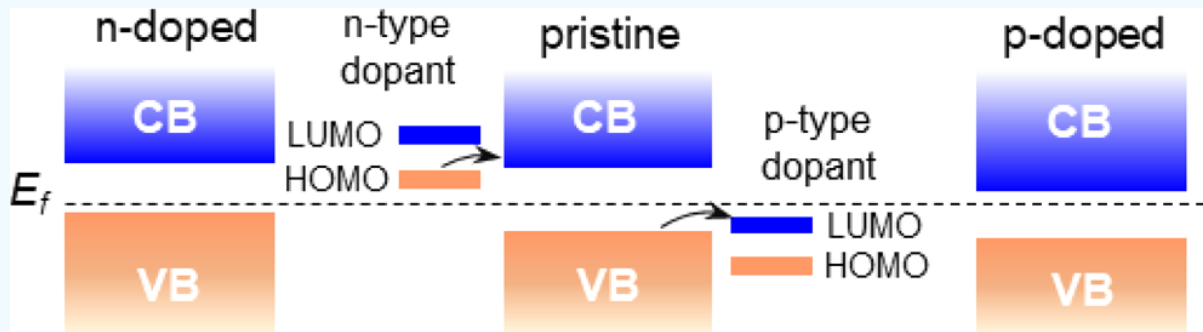
Simplicity, compatibility with wireless transmission, nonline-of-sight detection, possibility of continuous monitoring, portability, and compatibility with standard electronic equipment

Electrical Sensor

Semiconductors tend to increase their conductance with higher temperature. Resistance change can be used to *determine temperature*. The well-known semiconductor thermistors react sensitively to small temperature differences.



Conductivity Change with Doping Density



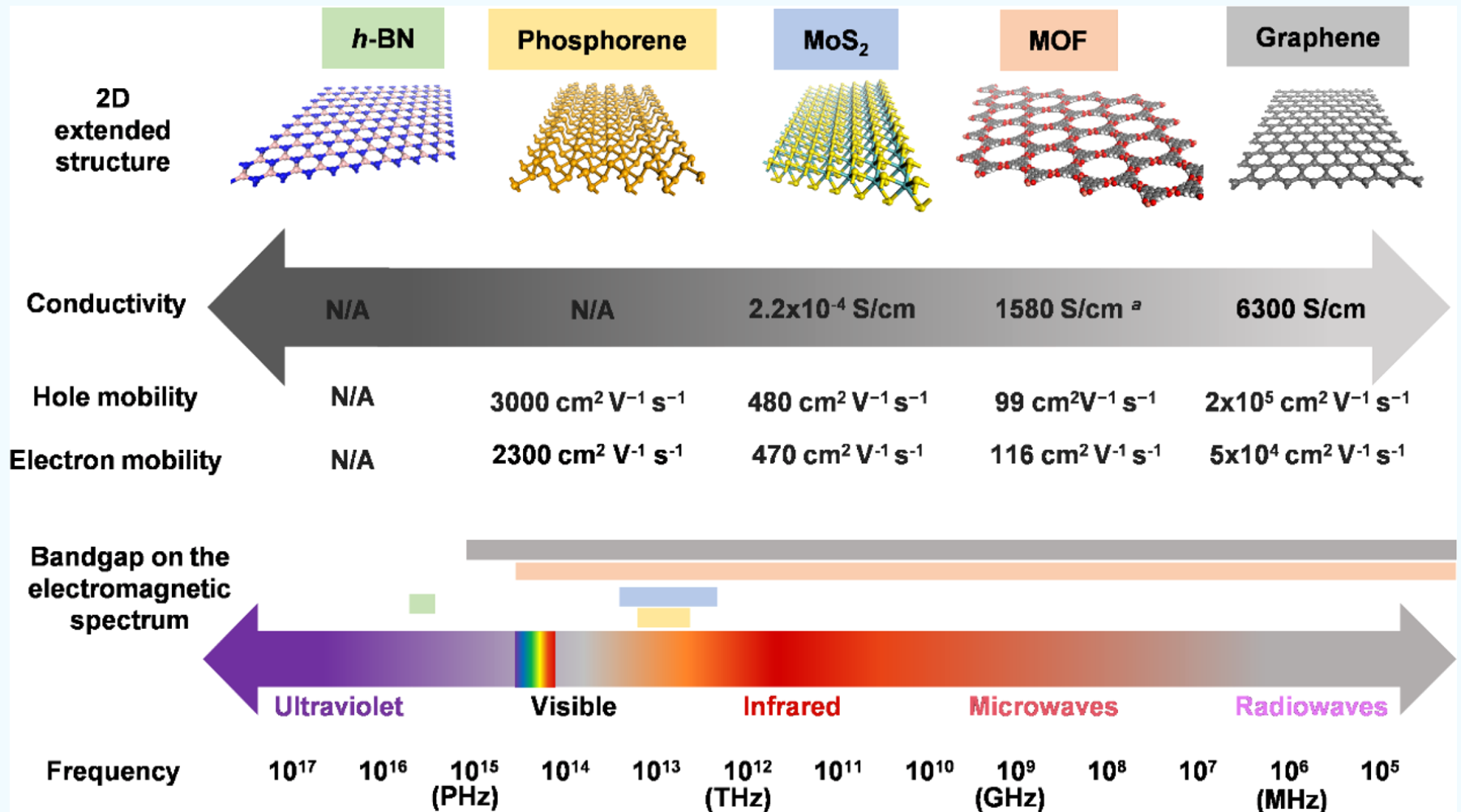
The analyte can act either as an n-type dopant or p-type dopant resulting in augmentation or compensation of the *doping charge density* thus alters the conductivity. Also, *proximity of analyte to the doped-materials* can proportionally vary the conductivity.

Modulation of Schottky Barrier

Metal– semiconductor junctions are involved in many types of electronic devices.

Formation of Dipole and Interfacial Layer

Electrical Sensor



This wide range of properties including tuneable conductivity, large surface to volume ration, flexibility, functionalization ability exhibited by these diverse 2D materials is the key to their success in sensing.